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Developments in Constrained Control Using Reference Governors

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Abstract: The paper provides an overview of reference governors, command governors and extended command governors, which are add-on schemes for enforcing pointwise-in-time state and control constraints in set-point tracking problems. It emphasizes recent developments based on the authors work and a new strategy for implementing command governors as they apply to nonlinear systems. Comments on future directions in reference governor research are made. A list of references is included.

Keywords: Control of systems with constraints; Reference governor; Command governor; Extended command governor.

1. INTRODUCTION

Reference governors are *add-on* schemes for enforcing pointwise-in-time state and control constraints by modifying set-point commands to a closed-loop system. See Figure 1. In this paper, we use the term *reference governor* to refer to several such schemes, including scalar and vector conventional reference governors, command governors and extended command governors. The common feature of these schemes is that they, whenever possible, preserve the response of the closed loop system designed by conventional control techniques. In addition, reference governors have computationally efficient implementations for both linear and nonlinear systems with disturbances and parameter uncertainties.

Reference governors were first proposed as continuous-time algorithms in [23] and subsequently in a discrete-time framework in [14, 17]. Alternative formulations and extensions to linear systems with disturbances have appeared in [3, 4, 15], and references therein. Parameter governors [32] and extended command governors [18] have been proposed as generalizations of the reference governor. Approaches to reference governor design for nonlinear systems exist (see [1, 16, 36, 19, 7] and references therein), where the use of on-line simulations or sub-level sets of Lyapunov functions is advocated to guard against constraint violation.

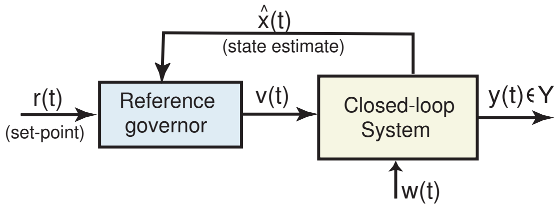

Fig. 1. Reference governor applied to closed-loop (Plant + Controller) system with constraints.

In this paper we survey several recent reference governor results, along with several applications. We then discuss several directions for future research.

2. REFERENCE GOVERNORS, COMMAND GOVERNORS AND EXTENDED COMMAND GOVERNORS

The basic approach can be described for a class of discrete-time linear systems with set-bounded disturbance inputs,

$$\begin{aligned} x(t+1) &= Ax(t) + Bv(t) + B_w w(t), \\ y(t) &= Cx(t) + Dv(t) + D_w w(t), \end{aligned} \tag{1}$$

where x is an n-vector state, v is an m-vector input, w is an l-vector disturbance, and y is a p-vector system output. The disturbance $w(t)$ is set-bounded in a given set W, i.e., $w(t) \in W$ for all $t \in Z^+$, where Z^+ denotes the set of non-negative integers.

In the applications of reference governors, (1) normally represents the closed-loop system and reflects the combined *closed-loop* dynamics of the plant and of the controller. These dynamics are thus asymptotically stable, and A is a Schur matrix (all eigenvalues are in the interior of the unit circle).

The state and control constraints on closed-loop variables, $y(t)$, are expressed as,

$$y(t) \in Y \text{ for all } t \in Z^+, \tag{2}$$

where Y is a prescribed set. Note that since (1) is a model of the closed-loop system, (2) can represent constraints on either state or control variables.

The conventional scalar and vector reference governors, command governors and extended command governors generate $v(t)$ in Figure 1 based on the current value of the reference, $r(t)$, the state estimate $x(t)$, and, in the case of the extended command governor, based also on an $\bar{n}$-vector state $\bar{x}$, of a supplementary, fictitious dynamic system. A detailed overview of these schemes is now provided. Some technical details are omitted. For example, under reasonable conditions, all of the governors have finite settling time in response to constant commands.

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2.1 Scalar reference governor (SRG)

The SRG uses the following update [14, 17, 15]:

$$v(t) = v(t-1) + \kappa(t)(r(t) - v(t-1)). \quad (3)$$

In (3), $\kappa(t)$ is a scalar parameter, maximized subject to the constraints that $0 \le \kappa(t) \le 1$ and $(v(t), x(t)) \in \tilde{O}_\infty$. The set $\tilde{O}_\infty$ is a finitely-determined inner approximation of the set of all states, $x(t)$, and constant inputs, $v(t+k) = v(t)$, such that for all disturbances satisfying $w(t+k) \in W,\ k \ge 0$, the subsequent response satisfies the constraints for all times: $y(t+k) \in Y$ for $k = 0, 1, \ldots, \infty$, and for $w(t+k) \in W,\ k \ge 0$. Procedures for computing $\tilde{O}_\infty$ are detailed in [13] and [31]. If the constraints imposed by $y \in Y$ are defined by linear inequalities, $\tilde{O}_\infty$ is a polyhedron represented by linear inequalities,

$$\tilde{O}_\infty = \{(x(0), v) :\ H_x x(0) + H_v v \le s\}. \quad (4)$$

Since only the scalar parameter $\kappa(t)$ is optimized on-line, the computational complexity of this approach is minimal and, in fact, the optimization problem is explicitly solvable [15]. If the system model or the constraints change, H_x, H_v and s can be computed on-line.

If $\kappa(t) = 1$, then $v(t) = r(t)$ and the reference governor does not interfere with the operation of the system. If a potential for constraint violation exists, the value of $\kappa(t)$ is decreased by the reference governor. *Recursive feasibility* is maintained at each time step: the value of $\kappa(t) = 0$ remains a feasible solution of the above optimization problem provided it is feasible at the initial time. Response properties of the reference governor, including conditions for the finite-time convergence of $v(t)$ to $r(t)$, are detailed in references [17, 15]. Essentially, if $r(t)$ for $t \ge t_0$ remains constant or varies in a sufficient small neighborhood of a constant value, then $v(t)$ converges to $r(t)$ if $r(t)$ is steady-state constraint admissible. Finally, we note that the implementation of the reference governor based on non-positively invariant subsets of $\tilde{O}_\infty$ is also possible [15].

As an illustration, consider a double integrator plant, $\dot{x}_1 = x_2$, $\dot{x}_2 = u$. The system is converted to discrete-time assuming a sampling period of $T_s = 0.1$ sec, and a nominal controller is designed using LQ techniques as $u = -0.917(x_1 - v) - 1.636x_2$, where v is a set-point for x_1. Suppose the state and control constraints are imposed as $|x_1| \le 1$, $|x_2| \le 0.1$, and $|u| \le 0.1$. The nominal response to a large command $v = r = 0.5$ from $x_1(0) = x_2(0) = 0$ results in maximum excursion of x_2 of about 0.2 and u of about 0.46, both exceeding the imposed limits. To avoid detuning the nominal controller (and thus compromising the response for small commands that do not cause constraint violation), we use SRG to govern v. The $\tilde{O}_\infty$ is a polytope defined by 36 linear inequalities. The closed-loop responses of the system to the command $r = 0.5$, after augmentation with SRG, are given in Fig. 2 and compared to the unconstrained response. The effect of the reference governor is simply to slow down the response.

2.2 Vector Reference Governor (VRG)

The VRG approach is an alternative to (3) when $m > 1$. It uses a diagonal matrix $\mathbf{K}(t)$ in place of a scalar $\kappa(t)$ to decouple the governing of different channels [17]:

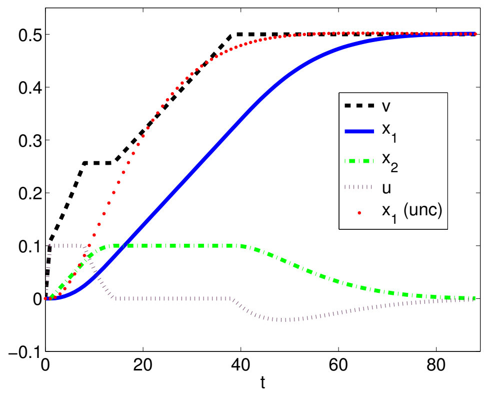

Fig. 2. The closed-loop response of the double integrator plant with the reference governor and the unconstrained response.

$$v(t) = v(t-1) + \mathbf{K}(t)(r(t) - v(t-1)), \quad (5)$$

where $\mathbf{K}(t) = \mathrm{diag}(\kappa_i(t))$. The values of $\kappa_i(t)$, $i = 1, \cdots, n$, are chosen to minimize $(v(t) - r(t))^{\mathrm{T}} Q (v(t) - r(t))$, where $Q = Q^{\mathrm{T}} > 0$ with $v(t)$ given by (5), subject to the constraints $0 \le \kappa_i(t) \le 1$, $i = 1, \cdots, n$, and $(v(t), x(t)) \in \tilde{O}_\infty$. This optimization problem can be solved online using quadratic programming techniques. Online use of conventional iterative techniques can be avoided by using explicit multi-parametric quadratic programming [6, 27]. The VRG is superior to the SRG in that it offers more flexibility in the choice of $v(t)$. In applications, it can provide faster response than the SRG.

2.3 Command Governor (CG) and Extended Command Governor (ECG)

The command governor and extended command governor approaches were proposed in [4, 8, 18] and references therein. The simplest variant of the command governor is similar to the VRG. In it, a cost function,

$$J = ||v(t) - r(t)||_Q^2 = (v(t) - r(t))^{\mathrm{T}} Q (v(t) - r(t)),$$

where $Q = Q^{\mathrm{T}} > 0$, is minimized with respect to $v(t)$ subject to the constraint that $(v(t), x(t)) \in \tilde{O}_\infty$.

The extended command governor approach of [18] is a predictive control strategy that generates $v(t)$ according to,

$$v(t) = \bar{C}\bar{x}(t) + \rho(t), \quad (6)$$

where, over the prediction horizon, the fictitious state, $\bar{x}(t) \in R^{\bar{n}}$, and $\rho(t)$ evolve as,

$$\bar{x}(t+k+1) = \bar{A}\bar{x}(t+k),\ k \ge 0,$$
$$\rho(t+k) = \rho(t). \quad (7)$$

The values of $\rho(t)$ and $\bar{x}(t)$ are optimized using a quadratic cost function,

$$J = \frac{1}{2}||\rho(t) - r(t)||_Q^2 + \frac{1}{2}||\bar{x}(t)||_P^2,$$

with $P = P^{\mathrm{T}} > 0$, satisfying $\bar{A}^{\mathrm{T}} P \bar{A} - P < 0$, subject to the constraint that $(\rho(t), \bar{x}(t), x(t)) \in \check{O}_\infty$. The set $\check{O}_\infty$ is a finitely determined inner approximation to the set of all $(\rho(t), \bar{x}(t), x(t))$ that do not induce subsequent constraint violation when the input sequence $v(t+k)$ is determined by the fictitious dynamics per (6) and (7). Without the fictitious states, i.e., when $\bar{n} = 0$, $\check{O}_\infty = \tilde{O}_\infty$, the ECG becomes the simple command governor [18]. The optimization problem can be solved online using conventional quadratic programming techniques. Again iterative procedures can be avoided by using explicit multi-parametric quadratic programming [6].

Various choices of $\bar{A}$ and $\bar{C}$ can be made. The shift sequences used in [18] are generated by the fictitious dynamics with,

$$\bar{A} = \begin{bmatrix} 0 & I_m & 0 & 0 & \cdots \\ 0 & 0 & I_m & 0 & \cdots \\ 0 & 0 & 0 & I_m & \cdots \\ 0 & 0 & 0 & 0 & \cdots \\ \vdots & \vdots & \vdots & \vdots & \ddots \end{bmatrix} \tag{8}$$

$$\bar{C} = [I_m \ 0 \ 0 \ 0 \ \cdots],$$

where I_m is an $m \times m$ identity matrix. Another approach [21], motivated by [38], uses Laguerre sequences. These sequences possess orthogonality properties and are generated by the fictitious dynamics with,

$$\bar{A} = \begin{bmatrix} \alpha I_m & \beta I_m & -\alpha\beta I_m & \alpha^2\beta I_m & \cdots \\ 0 & \alpha I_m & \beta I_m & -\alpha\beta I_m & \cdots \\ 0 & 0 & \alpha I_m & \beta I_m & \cdots \\ 0 & 0 & 0 & \alpha I_m & \cdots \\ \vdots & \vdots & \vdots & \vdots & \ddots \end{bmatrix} \tag{9}$$

$$\bar{C} = \sqrt{\beta}\left[I_m \ -\alpha I_m \ \alpha^2 I_m \ -\alpha^3 I_m \ \cdots \ (-\alpha)^{N-1} I_m\right],$$

where $\beta = 1 - \alpha^2$, and $0 \le \alpha \le 1$ is a selectable parameter that corresponds to the time-constant of the fictitious dynamics. Note that with the choice of $\alpha = 0$, (9) coincides with the shift register considered in [18]. The advantage of the ECG is that it can produce larger domain of attraction for $x(0)$ than is possible using the SRG, VRG, or CG.

2.4 Treatment of nonlinear systems

Approaches to reference governor design for nonlinear systems include [1, 16, 19, 7, 36], where the use of on-line simulations or sub-level sets of Lyapunov functions to guard against constraint violation is advocated. An incremental step reference governor strategy with one update of fixed magnitude per time step has been investigated in [40].

3. RECENT DEVELOPMENTS IN REFERENCE GOVERNORS

Several recent developments in the reference governor theory are now summarized.

3.1 Reference governors for linear systems with constraints expressed by nonlinear functions

Consider the case when the system dynamics (1) are linear, $W = \{0\}$, and the output constraints are specified by *nonlinear* functional inequalities. Specifically, Y is not polyhedral. It is given by

$$Y = \{y : \ h_i(y) \le 0, \ i = 1, \ldots, q\}, \tag{10}$$

where h_i are nonlinear functions. For some nonlinear systems it is possible by nonlinear feedback linearization [26] to render the closed loop system in Figure 1 linear. But then there is no way of making the resulting Y polyhedral. We note that the existing reference governor results in [17, 15] for system (1) with constraints (10) hold for any compact, convex set Y with $0 \in intY$. When Y is not polyhedral, it often can be approximated by a polyhedron. Such approximations, especially for multi-dimensional outputs, may not be easy to obtain or they may not be sufficiently simple and accurate.

The approach taken in [20] is to avoid the use of polyhedral approximations of Y and computation of $\tilde{O}_\infty$. Instead the linear model (1) is used to predict $y(t)$ and treat directly the violations of $h_i(y(t+k)) \le 0$ for $k > 0$. The results in [16] still apply in that, under reasonable assumptions, there exists a k^* such that it is only necessary to consider $0 \le k \le k^*$.

To predict the state and output of (1) at the time, $t + k$, given the state, $x(t)$, at time, t, and $v(t + k) = v(t)$ for $k \ge 0$, let

$$\Psi \triangleq (I - A)^{-1}B, \ \Gamma_x^k \triangleq CA^k, \ \Gamma_v^k \triangleq (C - \Gamma_x^k)\Psi + D. \tag{11}$$

Then the predicted output k steps ahead of the current time t is given by,

$$y(t + k|t) = \Gamma_x^k x(t) + \Gamma_v^k v(t). \tag{12}$$

With $v(t)$ given by (3),

$$y(t+k|t) = \Gamma_x^k x(t) + \Gamma_v^k v(t-1) + \kappa(t)\Gamma_v^k(r(t) - v(t-1)). \tag{13}$$

The constraints on $\kappa(t)$ are imposed by,

$$h_i(y(t + k|t) \le 0 \text{ for } i = 1, \ldots, q \text{ and } 0 \le k \le k^*. \tag{14}$$

3.2 Convex constraint functions

Suppose that h_i, $i = 1, \ldots, q$, in (10) are convex functions. By convexity of h_i and linearity of $y(t + k|k)$ in $\kappa(t)$, it follows that $h_i(y(t + k|t))$ in (14) is a convex function of $\kappa(t) \in [0, 1]$. In [20], we demonstrate that $\kappa(t) = \kappa_{\max}(t)$ where $\kappa_{\max}(t)$ is determined by a simple bisection procedure.

Further simplifications in calculating K_i^k occur if h_i are convex *quadratic* constraints of the form,

$$h_i(y) = y^{\mathrm{T}}\tilde{Q}_i y + \tilde{S}_i y + \tilde{C}_i \le 0, \tag{15}$$

where $\tilde{Q}_i = \tilde{Q}_i^{\mathrm{T}} \ge 0$. In this case it is not necessary to compute $\kappa_{\max}(t)$; the computations of $\kappa_{\max}(t)$ follow from the solution of a quadratic equation and are given by an explicit formula. See [20] for details.

3.3 Concave and MLD constraints

In [20], results and computational procedures are developed for several other special cases of nonlinear constraints. These include concave constraints, and "if-then" type Mixed Logical Dynamic (MLD) constraints.

3.4 Reduced order reference governors

The reduced order reference governor for systems with states decomposable into "slow" and "fast" states can be based on the reduced order model for "slow" states, provided constraints are tightened to ensure that the contribution of fast states does not cause constraint violation. The underlying ideas and results are briefly explained for the system (1) decomposed into slow and fast subsystems. See [22] for further details.

For simplicity it is assumed that $W = \{0\}$. After appropriate state transformations, the decomposed system has the following state space realization,

$$\begin{bmatrix} x_s(t+1) \\ x_f(t+1) \end{bmatrix} = \begin{bmatrix} A_s & 0 \\ 0 & A_f \end{bmatrix} \begin{bmatrix} x_s(t) \\ x_f(t) \end{bmatrix} + \begin{bmatrix} B_s \\ B_f \end{bmatrix} v(t), \quad (16)$$

$$y(t) = [C_s \ C_f] \begin{bmatrix} x_s(t) \\ x_f(t) \end{bmatrix} + Dv(t), \quad (17)$$

where $x_s(t) \in \mathbb{R}^{n_s}$ and $x_f(t) \in \mathbb{R}^{n_f}$ are the slow and fast vector states, respectively, $n_s + n_f = n$, and the matrices are appropriately sized. As (16)-(17) typically represents a closed-loop system, we assume that the matrices A_s and A_f are exponentially stable and eigenvalues of A_f are significantly smaller in magnitude than those of A_s.

Let $\Gamma_f = (I_{n_f} - A_f)^{-1} B_f$. Then the deviation of the vector of fast states, $x_f(t)$, from their predicted steady-state value over one time period, is denoted b,

$$e_f(t) = x_f(t) - \Gamma_f v(t-1). \quad (18)$$

The reference governor design is based on the reduced order system model, representing the dynamics of the slow states and steady-state values of fast states,

$$x_s(t+1) = A_s x_s(t) + B_s v(t), \quad (19)$$

$$y_r(t) = C_s x_s(t) + (C_f \Gamma_f + D) v(t). \quad (20)$$

We show in [22] that if $\Delta v(t) = v(t) - v(t-1)$ and $v(t+k) = v(t)$ for $k \geq 0$, then for all $k > 0$,

$$e_f(t+k) = A_f^k e_f(t) - A_f^k \Gamma_f \Delta v(t). \quad (21)$$

In order for the reference governor based on the reduced order model, (19)-(20), to enforce the constraints for the full order model, we tighten the constraints. Specifically, we introduce two "error sets" $E_x \subset \mathbb{R}^{n_f}$ and $E_y \subset \mathbb{R}^p$. We require that E_x be A_f-invariant, i.e.,

$$A_f E_x \subseteq E_x, \quad (22)$$

and also satisfy the following inclusion,

$$C_f E_x \subseteq E_y. \quad (23)$$

The sets E_x and E_y are design parameters. The set E_x is intended to contain the deviations of fast variables from the steady-state, $e_f(t)$. The choice $E_y = C_f E_x$ is an allowed choice; however, choosing a simpler E_y can reduce the computational complexity of the reference governor.

Suppose $\Delta v(t)$ is constrained by

$$-A_f \Gamma_f \Delta v(t) \in E_x \sim A_f E_x. \quad (24)$$

and that

$$y_r(t+k) \in Y \sim E_y, \ \forall k \geq 0. \quad (25)$$

Then, for all $k \geq 0$,

$$e_f(t+k) \in E_x, \quad (26)$$

and the constraints on the output (17) are satisfied, *i.e.*

$$y(t+k) \in Y. \quad (27)$$

Here, $\sim$ denotes the P-difference of two sets [1]. Using this result, the reference governor design can be based on the reduced order model (19) and (20) with the constraints (24), (25). The constraint (24) can be re-written as a constraint on $\kappa(t)$,

$$-A_f \Gamma_f \kappa(t)(r(t) - v(t-1)) \in E_x \sim A_f E_x.$$

If $v(-1)$ is initialized so that $e_f(0) = x_f(0) - \Gamma_f v(-1) \in E_x$, the reference governor response properties [15] hold, specifically, the recursive feasibility of $\kappa(t) = 0$, the guaranteed constraint enforcement and the finite-time convergence for constant set-points.

If an observer is used to estimate the slow states, a similar approach can be used to handle the observer error. Define

$$\tilde{A} = \begin{bmatrix} A_s - LC_s & LC_f \\ 0 & A_f \end{bmatrix}, \ \tilde{B} = \begin{bmatrix} -LC_f \Gamma_f \\ -A_f \Gamma_f \end{bmatrix}, \quad (28)$$

where L is the gain of the reduced order Luenberger observer for *slow states* only. To simultaneously handle the observer error, $\tilde{e}_s(t)$, and fast state deviations we introduce two error sets, $\tilde{E}_x \subset \mathbb{R}^n$, $n = n_s + n_f$, and $\tilde{E}_y \subset \mathbb{R}^p$, then

$$\tilde{A}\tilde{E}_x \subseteq \tilde{E}_x, \quad (29)$$

$$C\tilde{E}_x \subseteq \tilde{E}_y. \quad (30)$$

It then can be shown that if $(\tilde{e}_s(t), e_f(t)) \in \tilde{E}_x$, $v(t+k) = v(t)$ for $k \geq 0$,

$$\hat{y}_r(t+k) \in Y \sim \tilde{E}_y, \ \text{ for } k \geq 0, \quad (31)$$

and

$$\tilde{B}\Delta v(t) \in \tilde{E}_x \sim \tilde{A}\tilde{E}_x. \quad (32)$$

imply that for all $k \geq 0$, $(\tilde{e}_s(t+k), e_f(t+k)) \in \tilde{E}_x$, and the constraints on the output (17) are enforced, *i.e.* $y(t+k) \in Y$.

Even though these results require the error set E_x (respectively, $\tilde{E}_x$) to be A_f-invariant (respectively, $\tilde{A}$-invariant), this set must be A_f-contractive (respectively, $\tilde{A}_f$-contractive) in order to avoid overly restrictive constraints. For instance, if $\tilde{A}\tilde{E}_x = \tilde{E}_x$, where $\tilde{E}_x$ is compact, then $\tilde{E}_x \sim \tilde{A}\tilde{E}_x = 0$, resulting in an overly restrictive constraint, $\tilde{B}\Delta v(t) \in 0$.

The condition (24) constrains the increments of $v(t)$. The approach of constraining the increments of $v(t)$ has been also used to handle the case of network communication induced delays [10, 11] (discussed next). This approach has been also used by the authors of [12] for sensorless control.

3.5 Network reference governor

Reference governor-based approaches for networked control have been proposed in [2, 9] and recently in [10, 11]. In [10, 11], the set-point commands, $v(t)$, are transmitted through a communication channel that has time delays. In a simple case, when the delay, $\delta(t)$, is a fraction of the reference governor update period, T_s, the effect of the delay on the state is shown in [10] to satisfy the following relation:

$$\begin{aligned} x(t+1) &= Ax(t) + Bv(t) + R(\delta(t))\Delta v(t), \\ \Delta v(t) &= v(t+1) - v(t), \end{aligned} \quad (33)$$

with

$$R(\delta) = -\int_0^{\delta} e^{A_c(T_s - \tau)} B_c d\tau,$$

[1] $A \sim B := \{a \in A : a + b \in A, \ \forall b \in B\}$. See [31].

and (A_c, B_c) being the continuous-time realization of the remote system. Consequently, the effective disturbance introduced by the delay in (33) is modulated by $\Delta v(t)$, i.e., change in Δv.

To overbound $R(\delta)\Delta v$, suppose a matrix P and a set of vertices $\{w_i,\ i = 1, \cdots, n_w\}$ is given such that $||P\Delta v||_\infty \leq 1$ implies $R(\tau)\Delta v \subset convh\{w_i,\ i = 1, \cdots, n_w\}$ for all $0 \leq \tau \leq T_s$, where *convh* denotes the convex hull. Then the reference governor algorithm that ensures constraint enforcement and finite time convergence properties has the following form [10],

$$\begin{gathered}\min_{v(t),\zeta} ||r(t) - v(t)||_2^2,\\ \text{subject to}\\ H_x\Big(Ax(t) + Bv(t) + H_v v(t)\Big) \leq s - \max_{i=1,\cdots,n_w} (H_x w_i)\zeta\\ \zeta \geq [P(v(t) - v(t-1)]_j,\\ \zeta \geq -[P(v(t) - v(t-1)]_j,\\ j = 1, \cdots, m,\end{gathered} \tag{34}$$

where H_x, H_v and s are matrices in the representation of $\tilde{O}_\infty = \{(x, v) : H_x x + H_v v \leq s\}$.

Several extensions of these results are developed in [10, 11]. They include [10] the treatment of the combined delays in feedback and feed-forward channels, the output measurement case and a longer (random, possibly unbounded) delay for which command acceptance/rejection logic is augmented on the remote system side. A further extension is developed in [11], where the delay is assumed to be slowly varying with a known bound on the time rate of change and known at the time instant the command is sent. The approach combines the Smith predictor based on the estimated delay value and additive disturbance to cover the effect of the delay uncertainty.

3.6 Command governor for nonlinear systems

In the Lyapunov function based framework of [16, 36] for nonlinear systems, the command governor determines $v(t)$ as a solution to the following constrained optimization problem,

$$\min \tfrac{1}{2}(r - v)^{\mathrm{T}} Q(r - v) \text{ with respect to } v, \tag{35}$$

subject to

$$V(v, x) \leq c_i,\ i = 1, \cdots, q, \tag{36}$$

where V denotes the Lyapunov function of the closed-loop system and c_i, $i = 1, \cdots, q$ are constants induced by various constraints. Specifically $V(v(t), x(t)) \leq \min_{i=1,\cdots,q} c_i$ implies that the subsequent response with $v(t+k|t) = v(t)$ satisfies the imposed constraints $y(t) \in Y$.

We present a simple strategy for solving (35)-(36) online. Considering only the ith constraint in (36), the KKT conditions imply,

$$Q(r - v) + \lambda_i \frac{\partial V}{\partial v}(v, x) = 0, \tag{37}$$

where $\lambda_i \geq 0$ is the scalar Lagrange multiplier. The equation (37) can be solved to produce $v = v^*(\lambda_i, x, r)$ as a function of λ_i, x and r. Then a root finding problem is solved to find a scalar value of $\lambda_i = \lambda_i^*$ such that

$$V(v^*(\lambda_i^*, x, r), x) = c_i. \tag{38}$$

Thus the problem (35)-(36) reduces to

$$\min J = \tfrac{1}{2}(r - v)^{\mathrm{T}} Q(r - v) \text{ with respect to } v, \tag{39}$$

subject to

$$v \in \{v^*(\lambda_i^*, x, r),\ i = 1, \cdots, q\}, \tag{40}$$

i.e., selecting the best out of q solutions.

The scalar root finding problem in (38) can be solved using bisections. Alternatively, the predictor-corrector algorithm based on a single iteration per time step of Newton's method can be used to update λ_i^* in response to changes in x and r. Let

$$\bar{V}(\lambda_i, x, r) = V(v^*(\lambda_i, x, r), x) - c_i.$$

The predictor-corrector algorithm with one Newton-type iteration per time step has the following form,

$$\lambda_i(t+1) = \lambda_i(t) + \left(\frac{\partial \bar{V}}{\partial \lambda_i}\right)^{-1} \Big(-\bar{V}(\lambda_i(t), x(t), r(t+1)) - \frac{\partial \bar{V}}{\partial x}(x(t+1) - x(t)) - \frac{\partial \bar{V}}{\partial r}(r(t+1) - r(t))\Big),$$

where the partial derivatives are evaluated at $x(t)$, $v(t)$ and $r(t+1)$.

To illustrate this approach, we consider a double integrator, $\ddot{x} = u$ with the control constraint $|u(t)| \leq 1$. The feedback law has the form $u = -\omega_n^2(x_1 - v) - 2\zeta\omega_n x_2$, $x_1 = x$, $x_2 = \dot{x}$, where $\omega_n = 1$ and $\zeta = 0.1$. The closed-loop system is discretized with the sampling period $T_s = 0.01$ sec. In this case, the use of $V(x, v) = \frac{1}{2}\omega_n^2(x_1 - v)^2 + \frac{1}{2}x_2^2$, with $q = 1$ and $c_1 = 1/(2\omega_n^2(1 + 4\zeta^2))$ ensures constraint enforcement. Simulation results in Figures 3-6 demonstrate that the approach of solving (38) approximately based on the predictor-corrector algorithm with one iteration per time step gives very close results to the exact solution of (39).

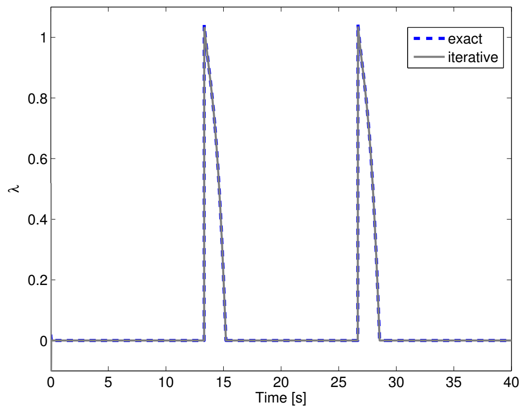

Fig. 3. The time histories of λ_1 for the exact and iterative in time predictor-corrector method.

4. APPLICATIONS

Some of the early references have considered the use of reference governors and similar schemes for flight control applications, see e.g., [17], [37], [39]. Several more recent applications of reference and extended command governors, based on the author's work, are discussed in what follows.

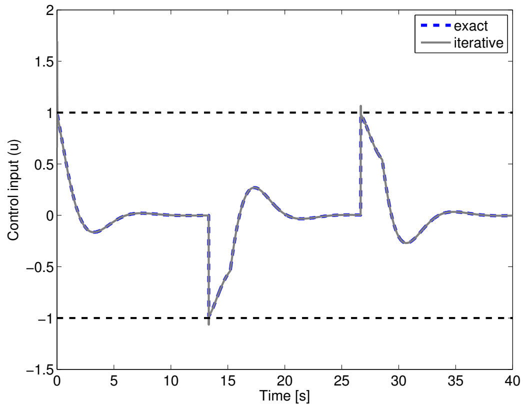

Fig. 4. The time histories of the control signal for the exact and iterative in time predictor-corrector method. The constraints are shown by the dashed lines. The constraint violation with the iterative in time predictor-corrector method is slight.

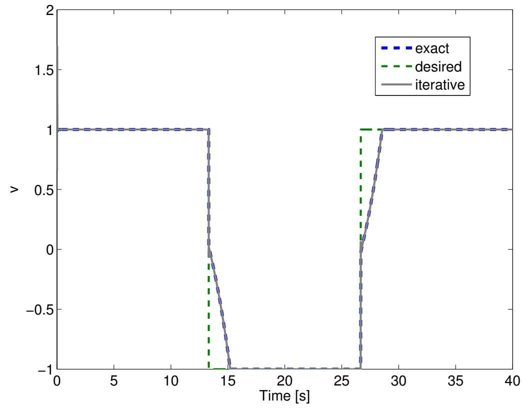

Fig. 5. The time histories of r and v for the exact and iterative in time predictor-corrector method.

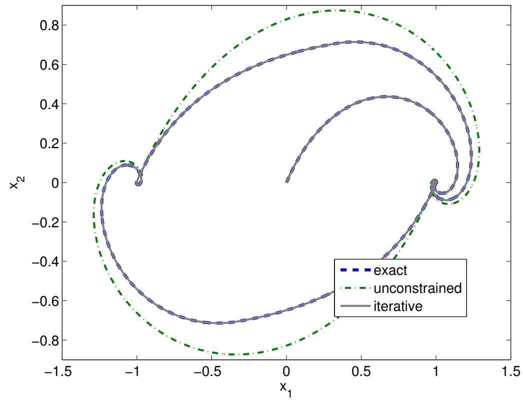

Fig. 6. The trajectory of x_1 and x_2 plane for the exact and iterative in time predictor corrector method.

4.1 Turbocharged automotive engines

As gasoline engines are downsized and turbocharged to improve fuel consumption, protecting the engine from violating the constraints without compromising engine response is becoming harder and requires systematic treatment. The surge constraint handling in turbocharged gasoline engines (see the schematic in Fig. 7) using reference governor techniques is addressed in [20], [21].

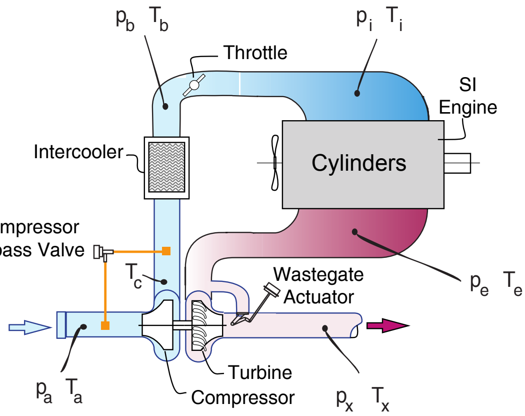

Fig. 7. Schematics of a turbocharged gasoline engine.

The application of the reduced order reference governor is motivated by different time scales of the engine model variables [20]. The model is order five with state variables: intake manifold pressure (kPa), boost pressure (kPa), exhaust manifold pressure (kPa), turbocharger speed (rpm), and wastegate flow (g/sec). The eigenvalues of the linearized continuous-time model are,

$$\{-2.39, -3.16, -24.3, -161, -259\},$$

suggesting that the dynamics can be decomposed into a second or a third order slow subsystem and, respectively, a third or a second order fast subsystem. The model has 2 outputs ($y(t)$): boost pressure (kPa), and compressor flow (g/sec)) that to avoid surge are constrained by an affine inequality as $y(t) \in Y$, where,

$$Y = \{y : Sy \leq s\}. \tag{41}$$

The reference governor is applied to modify the electronic throttle (ETC) command and wastegate command.

The linearized discrete-time model of the engine is first transformed into the form of (16)-(17), where the eigenvalues of A_f are fast and the eigenvalues of A_s are slow. Note that for this system, $D = 0$. We then proceed to define E_y by shrinking the constraint set, Y:

$$E_y = \{y : Sy \leq \varepsilon s\}, \tag{42}$$

where $\varepsilon > 0$ is a scalar. Therefore,

$$Y \sim E_y = \{y : Sy \leq (1 - \varepsilon)s\}. \tag{43}$$

Now, define, $E := \{x : SCx \leq \varepsilon s\}$. We construct a A_f-contractive set, E_x, with the maximum output admissible set algorithm [31], computing $O_\infty(\frac{1}{\lambda}A, C, \varepsilon Y)$, and obtaining a λ-contractive set with contraction parameter, λ. The responses of the system and the reference governor based on the linearized model for the different orders varying from 1 to 5 are shown in Figures 8-9. Consequently, the reduced order reference governor can be designed based on $n_s = 2$ without performance loss. The validation results based on the nonlinear model and the observer for unmeasured states are presented in [22].

We also note the related work [42, 41] on handling constraints in fuel cell applications. In these systems constraints are imposed to protect the fuel cell from oxygen starvation and air side compressor surge.

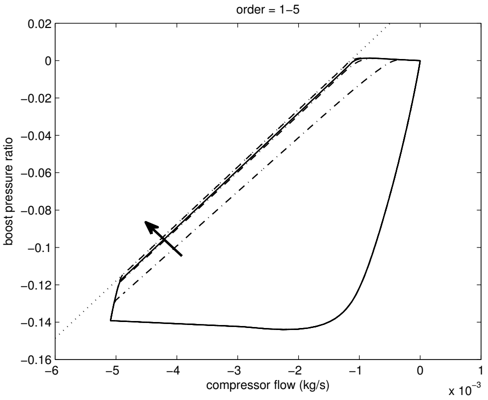

Fig. 8. The responses of a system of order varying from 1 to 5 with fixed $\varepsilon = 0.05$ plotted on a compressor map. The lowest two reduced order (dot-dashed) compared to the next two (solid) and the full-order (dashed) responses plotted against the compressor line (dotted). The arrows indicate the direction of increasing order.

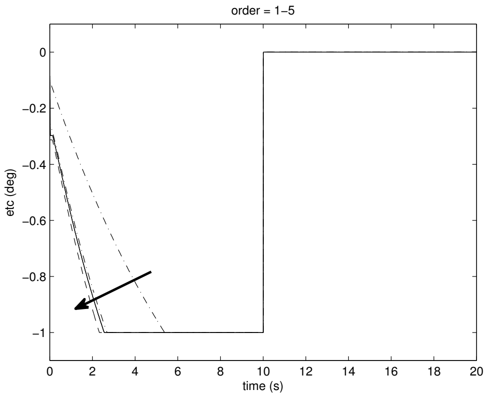

Fig. 9. The throttle responses of a system of order varying from 1 to 5 with fixed $\varepsilon = 0.05$. The lowest two reduced order (dot-dashed) compared to the next two (solid) and the full-order (dashed). The arrows indicate the direction of increasing order.

4.2 Aircraft gas turbine engines

In an aircraft engine, the surge can occur in the Fan, Low Pressure Compressor or High Pressure Compressor. In [28], the reference governor is applied to protect the engine against the violation of surge constraints. The reference governor modifies the set-point vector, consisting of Engine Pressure Ratio, Variable Stator Vane position and Varable Bleed Valve (VBV) position. In addition, the reference governor in [28] accommodates circumferential and radial inlet distortion uncertainties that can be caused by a high angle of attack, engine icing, bird ingestion, or the damage incurred during flight.

4.3 Spacecraft relative motion control

The application of the reference governor to spacecraft relative motion control is considered in [20]. In this problem, a *linear* Hill-Clohessy-Wiltshire model is used to represent the spacecraft relative motion dynamics. Several different types of constraints are considered. The reference governor handles nonlinear convex quadratic constraints on the thrust magnitude and on the Deputy spacecraft to approach the Chief spacecraft within the Line-of-Sight (LoS) cone emanating from the docking port. The reference governor also handles soft docking constraints that are of MLD type, and several other, affine constraints.

4.4 Vehicle dynamics and rollover protection

The application of the SRG and ECG to augment nominal steering angle during vehicle maneuvers so that vehicle constraints are satisfied is considered in [33]. The dynamics of vehicle yaw and roll motion are captured by a discrete-time model with four states (lateral velocity of Center of Gravity of the vehicle, yaw rate of the unsprung mass, roll rate of the sprung mass and roll angle of the sprung mass). The constraints on the load transfer ratio, which is the difference between the load on right tires minus the load on left tires divided by total weight of the vehicle, between -1 and 1, are enforced. As shown in [33], both SRG and ECG prevent rollover, while modifications of the vehicle trajectory are slight. The on-line computing effort associated with the SRG (explicitly solvable scalar optimization problem) is shown to be less than of the extended command governor (simple and explicitly solvable parametric quadratic program), while the domain of recoverable states of the extended command governor can be much larger. Further as Figures 10- 12 illustrate, ECG is robust to a mismatch between the model and actual system dynamics.

4.5 Electromagnetic actuators

Reference governors were applied to enforce a variety of constraints in electromagnetic actuators. See [25], [20], [36], [29], [30]. In particular, the constraint induced by the limited coil current leads to a constraint expressed by a concave nonlinear function, while the soft landing constraint is of MLD type. These types of constraints can be handled effectively using techniques in [20].

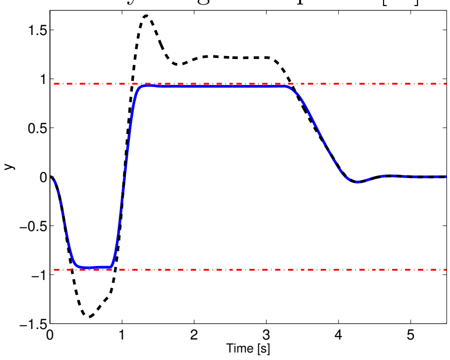

Fig. 10. The time histories of load transfer ratio $y(t)$ with ECG designed based on $v = 22.5$ m/sec model and $\tilde{O}_{\infty}(22.5)$ as terminal set (solid) and without ECG (dashed) for the vehicle maneuver at 30 m/sec. The constraints are shown by the dash-dotted lines.

5. RESEARCH TOPICS

While the subject of reference governors has been researched for over twenty years and blends into Modern Predictive Control, a variety of new research directions can be identified. We comment on two.

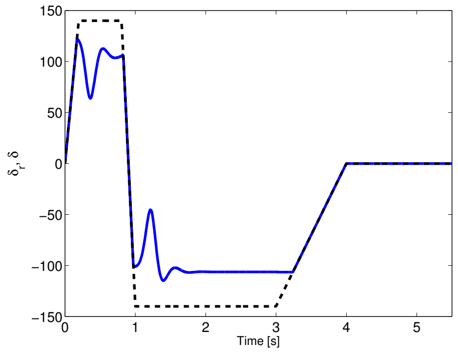

Fig. 11. The time histories of commanded steering angle $\delta_r(t)$ (dashed) and of $\delta(t)$ (solid) for the vehicle maneuver at 30 m/sec, where $\delta(t)$ is prescribed by the ECG designed based on $v = 22.5$ m/sec model and $\tilde{O}_\infty(22.5)$.

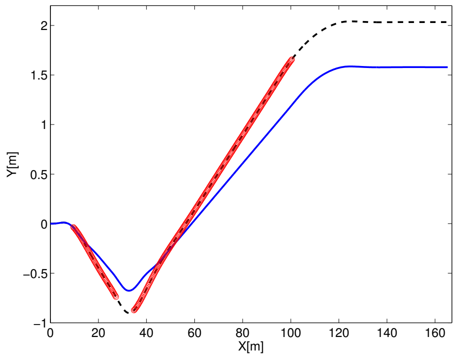

Fig. 12. Vehicle trajectory on the longitudinal-lateral plane with ECG (solid) and without ECG (dashed) for the vehicle maneuver at 30 m/sec, where $\delta(t)$ is prescribed by the ECG designed based on $v = 22.5$ m/sec model and $\tilde{O}_\infty(22.5)$. The points where the load transfer ratio constraint is violated without the reference governor are highlighted by circles.

5.1 Reference governor in the loop

Traditionally, reference and extended command governors modify the set-points to closed-loop systems. While the theory assumes that the set-points are given, in real systems the set-points may be adjusted by a human operator in response to external conditions. This dependence of set-points on external conditions can create a feedback loop encompassing the reference governor and nominal closed-loop system. A closely related situation occurs when the governor augments control signals at the *actuator command* level. This implementation of reference governors is appealing as the design and calibration of the nominal controller can be changed without the need to re-design the governor. The properties of the reference governor in the loop remain to be studied. It is interesting that the topic of error governors (see [13, 24] and references therein) that are related to reference governors in the loop, has received relatively little attention.

5.2 Reference governor for adaptive systems

Many applications require the treatment of plants with uncertain parameters. Several of the reference governor results (see e.g., [16, 41] and references therein) extend to the case when the system has uncertain set-bounded constant parameters. The application of the reference governors to systems with uncertain parameters being estimated online remains largely a future research topic. For instance, special assumptions are necessary in this case to guarantee recursive feasibility. A related case is of constraints varying in-time. While we have had success in treating specific examples, see e.g., [34], the theory remains largely to be developed.

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